Giustacchini 2017 CML — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

29/34 primary samples (≥29 cells) intermix on one continuous manifold with no per-patient islands (iLISI 8.67 of 23.39 ceiling, n_pcs=40 θ=10; radius-graph: 0 detached). K562 cell line (91 cells) scoped out — a detached, cycling clone reproducing none of the reference relationships.

Gate 2 — module geography → GO (non-branching)

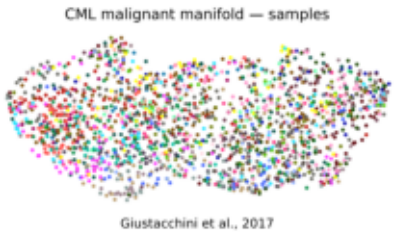
Single developmental axis: Gast/pSom overlap (r=−0.04, no competing arms); Crest an inverse terminal gradient opposing Gast (r=−0.88) and pSom (r=−0.37). Along pseudotime Gast falls (ρ=−0.96), Crest rises to a terminal readout (ρ=+0.79; crest-high decile mean pt=0.98).

Gate 3 — cross-repression → NA

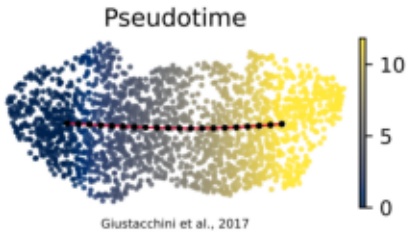
Single monotone axis — no bifurcation and no Gast/pSom co-expression hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

Manifold — samples



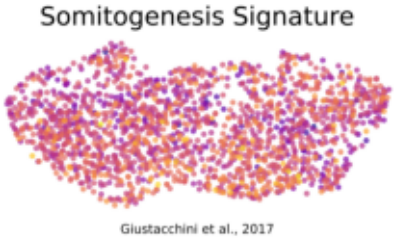
Pseudotime



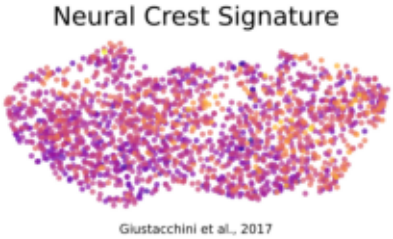
Gastrulation



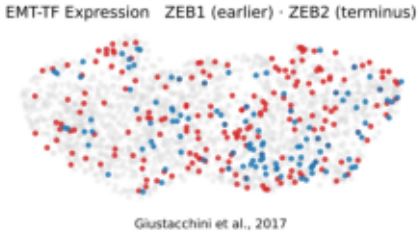
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Chronic myeloid leukemia (Giustacchini et al., 2017), SmartSeq2/HiSeq 2000 — 42 samples / 2,287 cells (master scope); malignant only: 2,055 load, 1,845 primary cells across 29 samples (≥29) enter the trajectory after the K562 cell line is scoped out. Non-branching Linear-pSom/Crest.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The primary CML malignant manifold is a single monotone gastrulation → terminal-neural-crest gradient rooted at the Gast/pSom-high, Crest-low origin; it lacks Puram's hub-and-spoke bifurcation (PPT tree: 2 tips, 0 forks), so Gate 3 has no hub to test. Crest peaks at the low-Gast terminus — no inversion. EMT-TFs ZEB1 (earlier) → ZEB2 (terminus); SNAI1/TWIST1/FOXC1/PRRX1 below the 3% floor.

Parameter choices

Min cells/sample	≥29 (keeps 29/34; smooth, 30→29 step)	**Scope**	primary tumour (K562 cell line dropped)
Root node	tip 0 = Gast/pSom-high, Crest-low	**Integration**	Harmony θ=10; n_pcs=40; PCA50; spread=1.3
Trajectory	scFates curve N=20, λ=0.3, μ=100	**Geometry**	Linear-pSom/Crest (non-branching)
EMT-TFs	ZEB1 (earlier) → ZEB2 (terminus); rest below 3% floor		